

Harmonization Reach Across Three Surveys

CPS, FoodAPS, and AHS scored against ACS: consolidated question-level results

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Numbers sources: CPS and FoodAPS values are the dedup-certified v1 results in `docs/validation/question_count`. AHS values are the v2 rollup in `v2/report/v2_report_numbers_collected.json` (ahs block). Reproducibility figures are from the v2 confirmation run (`v2/report/v2_results_report.qmd`). Nothing here is computed in prose except ratios of cited counts, each shown with numerator and denominator.

Scope

Three source surveys have been scored for harmonization against ACS as the anchor: CPS and FoodAPS in the v1 analysis (subsequently confirmed by the v2 reproducibility run), and AHS net-new in v2. The unit throughout is the unique question, not the question pair. A question counts as having a harmonization path if at least one of its pairs against ACS was rated F1 (direct recode) or F2 (statistical adjustment).

These three surveys carry different evidential weight and this report does not pool them. CPS and FoodAPS results are dual-generation validated: two independent model ensembles, one generation apart, reproduce the discard floor at 97 percent and agree on candidacy at 88 to 91 percent. AHS results come from a single model generation with no independent check, and are leads rather than findings. A pooled percentage across all three would put 744 unvalidated AHS questions into the same denominator as validated results and is deliberately not reported.

The funnel, per survey

Every rate depends on which denominator is chosen, so both are shown. The instrument denominator is every question the survey asks. The entered-pairing denominator is questions whose subtopics gave them any ACS territory to be compared against; the shortfall between the two is content with no ACS counterpart at all.

	CPS	FoodAPS	AHS
questions in instrument	211	462	744
entered pairing vs ACS	157	118	744
share of instrument entering	74.4%	25.5%	100%

	CPS	FoodAPS	AHS
produced no pairs (no shared subtopic)	0	0	263
at least one F1/F2 path	86	56	92
reach, of entered	54.8%	47.5%	12.4%
reach, of instrument	40.8%	12.1%	12.4%
provenance	v1, v2-confirmed	v1, v2-confirmed	v2 only, unvalidated

The pairing-stage accounting differs by design generation. In v1, questions with no shared ACS subtopic were filtered before pairing, so CPS and FoodAPS enter-pairing counts already exclude them. In v2, all in-map AHS questions entered and the 263 with no shared subtopic surface as an explicit bucket. The instrument-denominator rows are the comparable ones.

Tier composition of questions with a path

Each question is assigned its best tier across all of its pairs: F1 if any pair is a direct recode, otherwise F2 if any pair needs only statistical adjustment, otherwise F3.

best tier	CPS	FoodAPS	AHS
F1 direct recode	32	19	pending
F2 statistical adjustment	54	37	pending
F3 only (no path, but compared)	71	62	389
never compared (no shared subtopic)	54	344	263

The two AHS cells marked pending require a one-line count against `ahs_candidates.csv` on WORK (best feasibility per unique question over the 92 candidates); the rollup reports pair-level counts (71 F1 pairs, 59 F2 pairs) which are a different unit and are not substitutes. The CPS and FoodAPS never-compared rows are the instrument count minus entered-pairing count (211 minus 157; 462 minus 118).

Combined result for the validated pair (CPS + FoodAPS only)

For the two validated surveys, the combined question-level result is: 275 unique questions compared, 142 (51.6 percent) with at least one path, of which 51 (18.5 percent) are best-tier F1 and 91 (33.1 percent) best-tier F2; 133 (48.4 percent) are F3 throughout. On the ACS side, 48 of 115 ACS questions (41.7 percent) serve as targets for at least one source question, and 15 serve both CPS and FoodAPS simultaneously. No ACS-side participation figure exists yet for AHS.

What the three surveys say together

The pattern across the three is not one number but a gradient in how much of each survey lives on ACS territory. CPS is three-quarters ACS-adjacent and harmonizes at the highest rate. FoodAPS is a specialized instrument where only a quarter of the content can even be compared, but that quarter

harmonizes nearly as well as CPS. AHS enters in full yet reaches only 12 percent, because its bulk is housing-structure detail far deeper than anything ACS asks, and 263 of its questions have no ACS territory at all. Where AHS does touch ACS ground, the touch is close: its demographic questions reach 66 percent, in line with the validated surveys. The consistent story is that harmonization reach is a property of shared conceptual territory, not of the survey as a whole, and each instrument's unique content is exactly the enrichment it offers the others.

Caveats

AHS figures are single-generation and unvalidated; treat the 92 candidates as leads. The CPS and FoodAPS tier splits predate the v2 run and reflect v1 final classifications; the v2 confirmation showed v2 is the stricter rater on candidacy, so a v2-final tier split would likely be smaller. The never-compared rows for CPS and FoodAPS are derived by subtraction and inherit the v1 pairing-stage filter definition. Reach percentages use denominators shown in the same table; no percentage in this document mixes units.